CUBE.AI: The Ultimate Judge & Investor Q&A Bank

Project: Medical Image Segmentation & Agentic Radiology Assistant **Context:** Graduation Project Defense / Funding Pitch (ISF/Misr El Kheir)

💀 Section 1: The "Killer" Questions (Trap & Aggressive)

These are designed to test your confidence and honesty. Do not bluff.

Q1: "You are just students. What happens to this project when you get job offers from multinationals next month?"

- **Answer:** "We are not just students; we are co-founders. We have structured CUBE.AI as a spin-off with an IP agreement with Cairo University. We are committed to incorporated this startup post-graduation. The 80k EGP equipment remains with the lab under Dr. Meena Makary's supervision, ensuring the project's physical continuity regardless of our personal status."

Q2: "Why shouldn't I just buy a license from Siemens or GE? **They are safer."**

- **Answer:** "Siemens and GE offer closed ecosystems ('Black Boxes'). They do not allow you to train models on your *own* local data due to global regulatory rigidity. We offer **Localization**. We empower you to build AI that works for *Egyptian* patients and *your* specific scanner protocols. Safety comes from performance, and local models perform better locally."

Q3: "If your AI misses a tumor and a patient dies, who **goes to jail?"**

- **Answer:** "The Radiologist is always the 'Human-in-the-Loop'. CUBE.AI is legally a **Decision Support System (DSS)**, not a diagnostic device. Our UI forces the doctor to verify and approve every segmentation. We provide the *draft*; the doctor signs the *report*. Liability remains with the human professional."

Q4: "Aren't you just wrapping ChatGPT/LLMs and **calling it 'Medical AI'?"**

- **Answer:** "No. LLMs hallucinate. We use a **Multi-Agent Architecture**. The LLM (Planner Agent) acts only as the 'Router'—it understands the user's request. It then calls *deterministic*, clinically-validated tools (like nnU-Net or SAM) to do the actual medical work. We separate the 'Language' logic from the 'Medical' logic to prevent hallucinations."

Q5: "What is your 'Moat'? What stops a clever 2nd-year student from copying this in a weekend?"

- **Answer:** "The code is replicable; the **Workflow Integration** is not. Our moat is the **MCP (Model Context Protocol)** layer that integrates deeply with hospital PACS systems (DICOM standards) and the proprietary dataset of Egyptian pediatric cases we are building with our partners. Integration is the hard part, not the algorithm."

💰 Section 2: Financial & The "Ask" (80,000 EGP)

Q6: "Why **80,000 EGP? Why not 50,000?"**

- **Answer:** "The budget is strictly hardware-based. The price of an NVIDIA RTX 4090 Workstation is ~80k EGP. Anything less (e.g., RTX 3060) lacks the VRAM (24GB) required to load full 3D medical volumes. Lower specs would force us to downsample images, destroying diagnostic accuracy. 80k is the *minimum* physics-based requirement for 3D medical AI."

Q7: "Why not use Cloud (AWS/Google Colab)? It's cheaper upfront."

- **Answer:** "Two reasons: **Privacy and Bandwidth**.
  1. Egyptian Patient Data Protection laws prohibit uploading identified DICOM data to public servers abroad.
  2. Medical volumes are massive (500MB+ per patient). Uploading/downloading introduces unacceptable latency. On-premise computation is mandatory for clinical reality."

Q8: "If the dollar rate increases tomorrow, will this budget fail?"

- **Answer:** "We have built in a small contingency, but if hardware prices spike, we will downgrade non-essential peripherals (monitors/storage) to prioritize the GPU. The GPU is the engine; everything else is negotiable."

Q9: "What is your Customer Acquisition Cost (CAC)?"

- **Answer:** "Currently near zero. We are utilizing **Academic Partnerships** (Cairo University, CCHE 57357) for our pilot phase. We don't need a sales team yet; we need published clinical validation. Our 'Marketing' is 'Research Papers'."

Q10: "How much will you charge hospitals?"

- **Answer:** "We propose a **License + Maintenance** model. A one-time deployment fee (e.g., 50k-100k EGP) plus a 15% annual maintenance fee. This is significantly cheaper than the $100k+ annual licenses from foreign vendors."

🛠️ Section 3: Technical & Data

Q11: "Where are you getting your training **data?"**

- **Answer:** "We utilize two sources:
  1. **Public Benchmarks:** Medical Segmentation Decathlon (MSD) for base training.
  2. **Partner Data:** Anonymized pediatric neuroblastoma datasets from Cairo University Hospitals and CCHE (57357), secured via our supervisor Dr. Meena Makary."

Q12: "How do you handle 'Garbage In, Garbage Out' if a doctor trains a model badly?"

- **Answer:** "We implemented a **Data Hygiene Agent**. Before training starts, the system analyzes the inputs. If the images are low quality or the masks are inconsistent, it blocks the process and prompts the user to fix the data. We enforce quality assurance programmatically."

Q13: "What happens if the internet cuts off?"

- **Answer:** "CUBE.AI is designed for **Local Deployment**. The core segmentation engine runs entirely offline on the local workstation. Internet is only needed for updates, not for the daily clinical workflow."

Q14: "Why Neuroblastoma? It's a small market."

- **Answer:** "It is a **Proof of Capability**. Neuroblastoma is geometrically complex (wraps around vessels). If our AI can segment this, it can easily handle simpler tumors like Liver or Lung. Solving the 'Hardest Case' validates our tech for the wider market."

Q15: "What is the inference time?"

- **Answer:** "For interactive tools (SAM), it is real-time (<500ms). For full 3D volumetric automated segmentation, it takes **30-60 seconds** on the RTX 4090. This is perfectly acceptable for radiology reporting workflows."

🏥 Section 4: Clinical & Ethics

Q16: "Will this deskill junior radiologists?"

- **Answer:** "Drawing boundaries is a manual skill, not a cognitive one. We automate the tedious 'tracing,' allowing residents to focus on **Pathology and Diagnosis**. We shift their focus up the value chain."

Q17: "How **do you handle bias? (e.g., Model trained on adults used on kids)"**

- **Answer:** "That is exactly why CUBE.AI exists. Global models have this bias. Our 'Assisted Training' feature allows the hospital to fine-tune the model on *their* specific pediatric population, eliminating the bias inherent in generic global models."

Q18: "What if the radiologist gets lazy and just accepts the AI output without checking?"

- **Answer:** "This is 'Automation Bias.' To mitigate this, our UI requires **Active Confirmation**. The user must interact with the mask (e.g., verify key slices) before the 'Save to PACS' button becomes active."

👥 Section 5: Team & Execution

Q19: "You are all engineers. Who handles the business?"

- **Answer:** "We have split our team (as per the ISF application) into 'Innovators' (Tech) and 'Entrepreneurs' (Business). Team members like Meram and Shahd are focusing on the operational/business model, while Mostafa and Ibrahim focus on the tech. We also rely on Dr. Meena Makary for clinical strategy guidance."

Q20: "What is your timeline?"

- **Answer:**
  - **Now:** Prototype complete (running on laptop).
  - **Month 1 (Funded):** Buy Hardware (RTX 4090).
  - **Month 3:** Deploy Pilot at Cairo University Hospital.
  - **Month 6:** Publish Validation Paper & Apply for Seed Round."

🔮 Section 6: Future & Vision

Q21: "Can this expand beyond Radiology?"

- **Answer:** "Yes. The underlying technology (segmentation + agentic workflow) is applicable to **Pathology** (cell counting) and **Radiotherapy Planning**. However, we are laser-focused on Radiology for Phase 1."

Q22: "What is your exit **strategy?"**

- **Answer:** "Our goal is to become the standard 'AI Operating System' for Egyptian hospitals, then expand to MENA/GCC. Potential exit routes include acquisition by a regional healthcare provider (e.g., Alameda, Cleopatra Group) or a larger MedTech firm entering the African market."